

day/date

HW # 38

M. Abdullah Siddiqui

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CS-F

Problem 1:-

$$\Rightarrow A = \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix} \quad x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 + 3x_2 \\ 3x_1 + 5x_2 \end{bmatrix}$$

$$Q(x) = x_1^2 + 6x_1x_2 + 5x_2^2$$

$$\begin{vmatrix} 1-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = 0 \Rightarrow 5-6\lambda+\lambda^2-9=0$$

$$\lambda^2 - 6\lambda - 4 = 0$$

$$\lambda = 3 + \sqrt{13} \text{ or } \lambda = 3 - \sqrt{13}$$

Indefinite

$$\Rightarrow A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} x_1 - x_2 \\ -x_1 + x_2 \end{bmatrix}$$

$$Q(x) = x_1^2 - 2x_1x_2 + x_2^2$$

$$= (x_1 - x_2)^2$$

$$(1-\lambda)^2 - 1 = 0$$

Semi-positive definite

$$1 + \lambda^2 - 2\lambda = 0$$

$$\lambda = 0 \quad \lambda = 2$$

$$\Rightarrow A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \quad x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2x_1 + 3x_2 \\ 3x_1 + 5x_2 \end{bmatrix}$$

$$Q(x) = 2x_1^2 + 6x_1x_2 + 5x_2^2$$

$$(2-\lambda)(5-\lambda) - 9 = 0$$

$$\lambda^2 - 7\lambda + 1 = 0$$

Both eigen values &gt; 0

$$10 - 7\lambda + \lambda^2 - 9 = 0$$

Positive definite

$$\Rightarrow A = \begin{bmatrix} -1 & 2 \\ 2 & 8 \end{bmatrix} \quad Q(x) = -x_1^2 + 4x_1x_2 + 8x_2^2$$

$$(-1-\lambda)(8-\lambda) - 4 = 0$$

indefinite.

$$-8 + \lambda - 8\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 7\lambda - 12 = 0$$

VICKY



Problem 3:

$$z = 4xy$$

$$9x^2 + 4y^2 = 36$$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

$$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$

Change of variable  $x_1 = \frac{x}{2}$   $x_2 = \frac{y}{3}$

$$x = 2x_1 \quad y = 3x_2$$

Constraint:  $x_1^2 + x_2^2 = 1$

$$z = 4(2x_1)(3x_2)$$

$$z = 24x_1x_2$$

$$Q(x) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 0 & 12 \\ 12 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\lambda = \pm 12$$

$$\begin{bmatrix} -12 & 12 \\ 12 & -12 \end{bmatrix}$$

$$\begin{bmatrix} 12 & 12 & | & 0 \\ 12 & 12 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} -12 & 12 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{bmatrix} 12 & 12 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} x_2$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix} x_2$$

Max = 12  $x_1$  &  $x_2$  1 & 1

Min = -12  $x_1$  &  $x_2$  -1 & 1



Problem 4:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 4 \end{bmatrix}$$

Since  $x^T u_1 = 0$   $x_1 = 0$   
 $x_2^2 + x_3^2 = 1$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & 2 & 1 \\ 2 & 3-\lambda & 1 \\ 1 & 1 & 4-\lambda \end{vmatrix} = 0$$

$$(3-\lambda) \begin{vmatrix} 3-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} - 2 \begin{vmatrix} 2 & 1 \\ 1 & 4-\lambda \end{vmatrix} + 1 \begin{vmatrix} 2 & 3-\lambda \\ 1 & 1 \end{vmatrix} = 0$$

$$(3-\lambda)^2(4-\lambda) - (3-\lambda) - 2(8-2\lambda-1) + 1(2-3+\lambda) = 0$$

$$9 + \lambda^2 - 6\lambda(4-\lambda) - (3-\lambda) - 18 + 4\lambda + 2 + 2 - 3 + \lambda = 0$$

$$36 - 9\lambda + 4\lambda^2 - \lambda^3 - 24\lambda + 6\lambda^2 - 3 + \lambda - 18 + 4\lambda + 2 + 2 - 3 + \lambda = 0$$

$$-\lambda^3 + 10\lambda^2 - 27\lambda + 18 = 0$$

$$\lambda^3 - 10\lambda^2 + 27\lambda - 18 = 0$$

$$\lambda = 1 \quad \lambda = 3 \quad \lambda = 6$$

highest eigenvalue 6

$$\begin{bmatrix} -3 & 2 & 1 & | & 0 \\ 2 & -3 & 1 & | & 0 \\ 1 & 1 & -2 & | & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & | & 0 \\ 0 & 1 & -1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\begin{aligned} x_1 &= x_3 \\ x_2 &= x_3 \\ x_3 &= x_3 \end{aligned} \quad x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$



$$\left[ \begin{array}{ccc|c} 0 & 2 & 1 & 0 \\ 2 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right] \sim \left[ \begin{array}{ccc|c} 0 & 0 & 1/2 & 0 \\ 0 & 1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} x_1 = -1/2 x_3 \\ x_2 = -1/2 x_3 \\ x_3 = 2 \end{array}$$

$$x = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -1/\sqrt{6} \\ -1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

maximum value of  $x^T A x$  occurs along  $\lambda = 3$  the second max eigen value.